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Deep trench isolation structure and methods for fabrication thereof

Abstract

A Deep Trench Isolation (DTI) structure is disclosed. The DTI structures according to embodiments of the present disclosure include a composite passivation layer. In some embodiments, the composite passivation layer includes a hole accumulation layer and a defect repairing layer. The defect repairing layer is disposed between the hole accumulation layer and a semiconductor substrate in which the DTI structure is formed. The defect repairing layer reduces lattice defects in the interface, thus, reducing the density of interface trap (DIT) at the interface. Reduced density of interface trap facilitates strong hole accumulation, thus increasing the flat band voltage. In some embodiments, the hole accumulation layer according to the present disclosure is enhanced by an oxidization treatment.

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Background/Summary

BACKGROUND

(1) The semiconductor integrated circuit (IC) industry has experienced rapid growth. Technological advances in IC materials and design have produced generations of ICs where each generation has smaller and more complex circuits than the previous generation. However, these advances have increased the complexity of processing and manufacturing ICs and, for these advances to be realized, similar developments in IC processing and manufacturing are performed.

(2) Various common defects in image sensors, such as optical crosstalk, electrical crosstalk, dark current, and white pixels, become more serious as the image pixel sizes and the spacing between neighboring image pixels continues to shrink. Optical crosstalk refers to photon interference from neighboring pixels that degrades the light-sensing reliability and accuracy of the pixels. Dark current may be referred to the existence of pixel current when no actual illumination is present. In other words, the dark current is the current that flows through the photodiode when no photons are entering the photodiode. White pixels occur where an excessive amount of current leakage causes an abnormally high signal from the pixels,

(3) Deep trench isolation (DTI) structures are used to provide electrical and/or optical isolations between high voltage devices and image sensors. As the device dimension decreases, it is challenging to prevent leakage through current DTI structure design. For example, white pixel reduction becomes increasingly challenging for image sensors.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, according to the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

(2) FIG. **1** is a flow chart of a method for fabricating a semiconductor device including DTI structures according to embodiments of the present disclosure.

(3) FIGS. **2**, **2**A, **3**, **3**A, **4-7**, **7**A, **8-9**, **9**A, **10**, and **10**A schematically illustrate a semiconductor device at various stages of fabrication according to the method of FIG. **1**.

(4) FIG. **11** is schematic graph showing nitrogen concentration in a DTI structure according to embodiments of the present disclosure.

(5) FIG. **12** is a schematic graph showing oxygen concentration in a DTI structure according to embodiments of the present disclosure.

DETAILED DESCRIPTION

(6) The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

(7) Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “over,” “top,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative

terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

(8) A Deep Trench Isolation (DTI) structure in a semiconductor substrate and the method of forming the same are provided according to various embodiments. The intermediate stages of forming the DTI structure are illustrated according to some embodiments. Some variations of some embodiments are discussed.

(9) DTI structures according to embodiments of the present disclosure include a composite passivation layer. In some embodiments, the composite passivation layer includes a hole accumulation layer and a defect repairing layer. The defect repairing layer is disposed between the hole accumulation layer and a semiconductor substrate in which the DTI structure is formed. The defect repairing layer may be a nitrogen or hydrogen rich material. The defect repairing layer reduces lattice defects at the interface between the semiconductor substrate and the DTI structure, thus, reducing the density of interface trap (Dit) at the interface. Reduced density of interface trap facilitates strong hole accumulation at the interface, thus increasing the flat band voltage. In some embodiments, the hole accumulation layer according to the present disclosure is enhanced by an oxidization treatment.

(10) The DTI structure may be used for Backside Illumination (BSI) Complementary Metal-Oxide-Semiconductor (CMOS) image sensors or Front Side Illumination (FSI) CMOS image sensors, logic devices, and any suitable devices in which deep trench isolation are used. In an image sensing device, the defect repairing layer reduces lattice damages at the interface between the DTI structure and the semiconductor substrate, thus, reducing white pixel occurrence without using high temperature anneal.

(11) In image sensing devices, DTI structures may be formed on a front side of the semiconductor substrate with the transistors of the pixel elements or on the backside of the semiconductor substrate. Backside DTI structures that are fabricated after metallization process. To avoid damaging the prior formed metal features, backside DTI structures cannot be annealed at high temperature. The DTI design according to the present disclosure may not require an annealing process at a temperature higher than about 410° C., therefore is particularly beneficial to backside DTI structures.

(12) FIG. **1** is a flow chart of a method **100** for fabricating a semiconductor device including DTI structures according to embodiments of the present disclosure. FIGS. **2**, **2**A, **3**, **3**A, **4-7**, **7**A, **8-9**, **9**A, **10**, and **10**A schematically illustrate a semiconductor device **200** at various stages of fabrication according to the method **100**. In some embodiments, the semiconductor device **200** fabricated according to the method **100** includes BSI image sensing devices. It is understood that additional steps can be provided before, during, and/or after the method **100**, and some of the steps described can be replaced, eliminated, and/or moved around for additional embodiments of the method **100**.

(13) At operation **102** of the method **100**, an implantation process is performed to form a plurality of doped regions **206** in a semiconductor substrate **202**, as shown in FIGS. **2** and **2**A. FIG. **2** is a schematic cross-sectional view of the semiconductor device **200**. FIG. **2**A is a schematic top view of the semiconductor device **200**.

(14) According to some embodiments of the present disclosure, the semiconductor substrate **202** may be a crystalline silicon substrate. According to other embodiments of the present disclosure, the semiconductor substrate **202** includes an elementary semiconductor such as germanium; a compound semiconductor including silicon carbon, gallium arsenic, gallium phosphide, indium phosphide, indium arsenide, and/or indium antimonide; an alloy semiconductor including SiGe, GaAsP, AlInAs, AlGaAs, GaInAs, GaInP, and/or GaInAsP; or combinations thereof. Other substrates such as multi-layered or gradient substrates may also be used. The semiconductor

substrate **202** has a front surface **202***f* and a back surface **202***b*. In some embodiments, the front surface **202***f* and the back surface **202***b* may be on (100) or (001) surface planes.
(15) The plurality of doped regions **206** may be formed by a selective implantation process. A masking layer **208** may be deposited on the front surface **202***f* of the semiconductor substrate **202**. A patterning process, such as a photolithography process, is performed form a plurality of openings through the masking layer **208** and expose portions of the front surface **202***f* of the semiconductor substrate **202**. In some embodiments, the masking layer **208** may comprise photoresist or a nitride, for example silicon nitride (SiN), patterned using a photolithography process.
(16) The plurality of doped regions **206** are formed by an implantation process to drive implant dopants **210** into the semiconductor substrate **202** from the front surface **202***f*. As shown in FIG. **2**, the plurality of doped regions **206** are located within the semiconductor substrate **202** at a distance T**1** from the front surface **202***f*. The portion of the semiconductor substrate **202** above the plurality of doped regions **206** may be referred as a transistor region **204** because various transistors may be formed in and on the transistor region **204**. The plurality of doped regions **206** may have a thickness T**2**. The distance T**1** and thickness T**2** may be selected according to circuit design and achieved by adjusting bias applied to the semiconductor substrate **202** and flow density of the dopants **210** during implantation process.
(17) In some embodiments, the plurality of doped regions **206** are intended as light sensing regions for a plurality of pixels in an image sensing device. In some embodiments, the plurality of doped regions **206** may be doped by a n-type dopants and intent to be deep N-type pinned photodiodes (DNPPD) in the plurality of image sensors to be formed. The dopants **210** may include one or more n-type dopants, such as phosphorous, arsenic, antimony, or the like. In some embodiments, the plurality of doped regions **206** may include n-type dopants at a concentration in a range between about 1E15 atom/cm.sup.3 and about 1E20 atom/cm.sup.3.
(18) As shown in FIG. **2**A, the plurality of doped regions **206** are individual areas separated from one another. In some embodiments, each of the doped regions **206** may be form an array with gap regions **207** between neighboring doped regions. As discussed below, DTI structures may be formed in the gap regions **207** to electrically and/or optically isolate the dope regions **206**.
(19) DTI structures may include front side DTI structures and backside DTI structures. Front side DIT structures are formed from the front surface **202***f* during the front end of line (FEOL) processes. A plurality of trenches may be formed in the gap regions **207** of the semiconductor substrate **202** followed by deposition of an isolation layer or a passivation layer, such as one or more high-k dielectric films, on the exposed surfaces of the semiconductor substrate **202**. A high temperature annealing process may be performed to reduce density of defect traps at the interface between the semiconductor substrate **202** and the isolation layer or passivation layer. However, front side DTI structures have layout restrictions because front side DTI structures have to avoid areas for pixel transistors formed in and on the front surface **202***f* of the semiconductor substrate **202**. Backside DTI structures are formed from the back surface **206***b* of the semiconductor substrate **202** usually after FEOL processes and middle end of line (MEOL) processes are completed. Backside DTI structures are not constrained by the layout of pixel transistors formed in the transistor region **204**. However, defect traps in the interface between the semiconductor substrate **202** and the backside DTI structures can't be reduced by high temperature anneal to avoid damages to metallic features formed during FEOL and BEOL processes. The DTI structures according to the present disclosure may not need a high temperature anneal, therefore, may be formed from the backside to be both effective and design friendly.
(20) In some embodiments, the semiconductor substrate **202** may be a p-type substrate such that a p-n junction may be formed at the interface between the semiconductor substrate **202** and the DTI structure to be formed. Alternatively, a doping process may be performed to form deep p-wells (DPWs) **216** in the gap regions **207** of the semiconductor substrate **202**, as shown in FIGS. **3** and **3**A. DIT structures are subsequently formed in the DPWs **216**. The DPWs **216** may extend from the

front surface **202***f* to a depth so that the plurality of the doped regions **206** are surrounded by the DPWs **216** along the entire depth T**2**. In some embodiments, cell p-wells (CPWs) **217** may be formed at an upper portion of the DPWs **216**. One or more transistors may be formed on and in the CPWs **217**.

(21) The DPWs **216** and the CPWs **217** may be formed by a selective implantation process. A masking layer **212** may be deposited on the front surface **202***f* of the semiconductor substrate **202**. A patterning process, such as a photolithography process, is performed form a plurality of openings through the masking layer **212** and expose portions of the front surface **202***f* of the semiconductor substrate **202**. The DPWs **216** and the CPWs **217** may be formed by one or more implanting processes with one or more p-type dopants **214**. The p-type dopants **214** may include boron (B), aluminum (Al), and gallium (Ga). In some embodiments, the DPWs **216** may have a dopant concentration in a range from about 1E10 atom/cm.sup.3 to about 1E12 atom/cm.sup.3, for example, in a range of from about 2E11 atom/cm.sup.3 to about 7E11 atom/cm.sup.3. In some embodiments, the CPWs **217** may have a dopant concentration in a range from about 1E11 atom/cm.sup.3 to about 1E13 atom/cm.sup.3, for example, in a range of from about 1E12 atom/cm.sup.3 to about 6E12 atom/cm.sup.3. Even though FIG. **3** shows that the DPWs **216** and the CPWs **217** occupy the same areas, it should be noted that, according to design layout, the DPWs **216** and the CPWs **217** may occupy the same areas, different but overlapping areas, or different and not overlapping areas.

(22) At operation **104**, a plurality of device elements **222** are formed in and on the transistor region **204** of the semiconductor substrate **202**, as shown in FIG. **4**. The plurality of device elements **222** may be any devices, such as an image sensing device, a logic device, an input/output (I/O) device, a memory device. Each device element **222** may include one or more transistors, such as metal oxide semiconductor field effect transistors (MOSFET), complementary metal oxide semiconductor (CMOS) transistors, bipolar junction transistors (BJT), high voltage transistors, high frequency transistors, p-channel and/or n-channel field effect transistors (PFETs/NFETs), etc., diodes, and/or other applicable elements. In some embodiments, the device elements are formed in and on the semiconductor substrate **202** in a front-end-of-line (FEOL) process.

(23) In some embodiments, the plurality of device elements **222** are a plurality of pixel device for an image sensor. Each pixel device may include a transfer gate **218** which extends into the corresponding doped region **206**. Various transistors for a pixel device may be formed in the transistor region **204** of the semiconductor substrate **202** and the ILD layer **220**. For example, a pixel device may include a transfer transistor, a reset transistor, a source-follower transistor, and a select transistor. The pixel device may include other suitable transistors, such as a shutter gate transistor, a storage transfer transistor, or a combination thereof. Source/drain features for various transistors and shallow trench isolation (STI) may be formed in the transistor region **204** of the semiconductor substrate **202**. Gate structures for the various transistors may be formed in the ILD layer **220**.

(24) In operation **106**, an interconnect structure **224** is formed over the ILD layer **220**, as shown in FIG. **4**. The interconnect structure **224** includes multiple levels of conductive lines and conductive vias embedded multiple layers of dielectric materials to provide electrical paths to various the device elements **222** formed below. The dielectric material may be a low-k material, such as SiO.sub.x, SiO.sub.xC.sub.yH.sub.z, SiOCN, SiON, or SiO.sub.xC.sub.y, where x, y and z are integers or non-integers. In some embodiments, the interconnect structure **224** may include etch stop layers between levels of low-k dielectric material layers to facilitate patterning and formation of the conductive lines and conductive vias at different levels. The etch stop layers may be made of silicon carbide (SiC), silicon nitride (SixNy), silicon carbonitride (SiCN), silicon oxycarbide (SiOC), silicon oxycarbon nitride (SiOCN), tetraethoxysilane (TEOS) or another applicable material.

(25) The conductive lines and conductive vias may be made from one or more electrically

conductive materials, such as metal, metal alloy, metal nitride, or silicide. For example, the conductive lines and conductive vias are made from copper, aluminum, aluminum copper alloy, titanium, titanium nitride, tantalum, tantalum nitride, titanium silicon nitride, zirconium, gold, silver, cobalt, nickel, tungsten, tungsten nitride, tungsten silicon nitride, platinum, chromium, molybdenum, hafnium, iridium, other suitable conductive material, or a combination thereof.

(26) At operation **108**, deep isolation trenches **234** are formed in the DPWs **216** of the semiconductor substrate **202**, as shown in FIGS. **5** and **6**. In FIG. **5**, a carrier wafer **226** is attached to the interconnection structure **224** and the semiconductor substrate **202** is flipped over upside down for back side processing.

(27) A backside grinding may be performed to grind the back surface **202***b* to thin down the semiconductor substrate **202**. In some embodiments, the thickness of the semiconductor substrate **202** may be reduced to smaller than about 10 μm, or smaller than about 5 μm. In some embodiments, the semiconductor substrate **202** is grinded to expose the doped region **206**, resulting a back surface **206***b*, as shown in FIG. **5**.

(28) A masking layer **230** may be deposited on the back surface **206***b*. A patterning process, such as a photolithography process, is performed form a plurality of openings **232** through the masking layer **230** and expose the back surface **206***b* of the semiconductor substrate **202**. In some embodiments, the masking layer **230** may comprise photoresist or a nitride, such as SiN, patterned using a photolithography process.

(29) The openings **232** are aligned with the DPWs **216**. An etch process is performed to remove a portion of the DPWs **216** in the semiconductor substrate **202** and form the deep isolation trenches **234**. When viewed from the top, the deep isolation trenches **234** form a grid and surround the plurality of doped regions **206** for the pixel elements **222**.

(30) In some embodiments, an anisotropic etching process is performed so that sidewalls **234***w* of the deep isolation trenches **234** are straight and vertical, i.e., the sidewalls **234***w* are substantially perpendicular to the back surface **206***b*. In some embodiments, the deep isolation trenches **234** may also be slightly tapered, and hence the sidewalls **234***w* of the deep isolation trenches **234** are slightly tilted relative to the back surface **206***b*. For example, an angle α between the sidewall **234***w* and the back surface **206***b* may be greater than about 88 degrees and smaller than 90 degrees. In some embodiments, the deep isolation trenches **234** are formed within the DPWs **216** so that sidewalls **234***w* include p-type semiconductor material.

(31) In some embodiments, the deep isolation trenches **234** may have a depth D**1** in a range between about 0.5 μm and about 10 μm, and a width W**1** in a range between about 0.025 μm and about 0.3 μm. In some embodiments, an aspect ratio D**1**/W**1** of the deep isolation trenches **234** may be in a range between about 10 and 20. In some embodiments, the deep isolation trenches may extend through the thickness T**2** of the doped regions **206** to provide full coverage to the doped regions **206**. In other embodiments, the deep isolation trenches **234** may substantially cover the thickness T**2** of the doped regions **206**.

(32) In some embodiments, the etching process is performed through a dry etching method including, and not limited to, Inductively Coupled Plasma (ICP), Transformer Coupled Plasma (TCP), Electron Cyclotron Resonance (ECR), Reactive Ion Etch (RIE), and the like. The etching process may be performed using process gases including, fluorine-containing gases, such as SF.sub.6, CF.sub.4, CHF.sub.3, NF.sub.3, Chlorine-containing gases (such as Cl.sub.2), Br.sub.2, HBr, BCl.sub.3, and/or the like.

(33) At operation **110**, a defect repairing layer **236** is formed on the sidewalls **234***w* and the bottom **234***b* of the deep isolation trenches **234**, as shown in FIG. **6**. As discussed previously, the sidewalls **234***w* and the bottom **234***b* of the deep isolation trenches **234** include semiconductor material, such as p-type semiconductor materials of the semiconductor substrate **202** or the DPWs **216**.

(34) FIG. **7**A is a partial enlarged view of the area marked **7**A in FIG. **7**. As shown in FIG. **7**A, the defect repairing layer **236** has an interface **236***i* with the DPW **216** of the semiconductor substrate

202. Dangling bonds of the semiconductor element at the interface **236***i* would trap and fix charge, thus reduce hole accumulation in a hole accumulation layer. At the interface **236***i*, the defect repairing layer **236** is directly formed on the semiconductor materials on the sidewalls **234***w* and is configured to passivate dangling semiconductor bonds in the sidewalls **234***w* of the deep isolation trenches **234**. For example, when the semiconductor substrate **202** is a silicon substrate, the defect repairing layer **236** is configured to passivate the dangling silicon bonds at the sidewall **234***w*. In some embodiments, the density of interface trap (Dit) at the interface **236***i* is in a range between 3.6E11 and about 4.2E11. A typical density of interface defect at an interface between silicon and a high-k material, such as aluminum oxide, is about 1.3E12. Therefore, the defect repairing layer **236** reduces density of interface trap (Dit) at the interface **236***i* in a range between about 50% to about 70%.
(35) In some embodiments, the defect repairing layer **236** includes a nitrogen or hydrogen rich material, for example, a nitride, an oxynitride, a hydroxide. In some embodiments, the defect repairing layer **236** may be a nitride, an oxynitride, or a hydroxide of a metal or metal alloys. For example, the defect repairing layer **236** may be nitride, an oxynitride, or a hydroxide of aluminum, a transition metal, or alloys thereof. For example, the defect repairing layer **236** may be aluminum nitride (AlN), aluminum oxynitride (AlON), aluminum hydroxide (AlOH), hafnium oxynitride (HfON), Zirconium oxynitride (ZrON), titanium oxynitride (TiON), hafnium aluminum oxynitride (HfAlON), or the like.
(36) In some embodiments, the defect repairing layer **236** may be formed by a low temperature deposition process, such as a deposition process at a temperature lower than about 410° C., such as a plasma enhanced atomic layer deposition (PEALD), plasma enhanced chemical vapor deposition (PECVD), and any suitable process. The defect repairing layer **236** may be formed by using a precursor of a metal source, and a precursor of hydrogen or nitrogen.
(37) In some embodiments, the defect repairing layer **236** may include an aluminum nitride (AlN) layer deposited by PEALD, using an aluminum-containing precursor, such as trimethylaluminum (TMA), triethylaluminium (TEA), or other suitable chemical, and a nitrogen-containing precursor, such as ammonia (NH_3), tertiarybutylamine (TBAm), phenyl hydrazine, or other suitable chemical.
(38) In some embodiments, the defect repairing layer **236** is a nitrogen containing layer. In some embodiments, the defect repairing layer **236** includes a metal oxynitride deposited by alternatively flowing a metal containing precursor, an oxygen plasma, and an ammonia plasma with NH_3/N_2 or a nitrogen plasma.
(39) The defect repairing layer **236** may be conformally deposited on all exposed surfaces. In some embodiments, the defect repairing layer **236** has a thickness T**3** (shown in FIG. **10**A) in a range between about 1 angstrom and about 50 nm. If the thickness T**3** is thinner than 1 angstrom, the defect repairing layer **236** may not able to sufficiently reduce the density of interface defect. If the thickness T**3** is thicker than 50 nm, the defect repairing layer **236** may cause negative shift of flat band voltage without providing additional benefit of reducing density of interface defect.
(40) In operation **112**, a hole accumulation layer **238** is deposited on the defect repairing layer **236**, as shown in FIG. **7**. The hole accumulation layer **238** may include high-k material with high negativity to form hole accumulation in adjacent semiconductor, such as the semiconductor in the sidewall **234***w* of the deep isolation trenches **234**. In some embodiments, the hole accumulation layer **238** include one or more metal oxide with an areal oxygen density greater than the areal oxygen density in silicon oxide. In some embodiments, the hole accumulation layer **238** is formed of aluminum oxide (Al_2O_3), titanium oxide (TiO_2), hafnium oxide (HfO_2), tantalum oxide (Ta_2O_5), scandium oxide (Sc_2O_3), zirconium oxide (ZrO_2), magnesium oxide (MgO), lutetium oxide (Lu_2O_3), yttrium oxide (Y_2O_3), lanthanum oxide (La_2O_3), hafnium aluminum oxide (HfAlO), or the like, or a composite layer including more than one of these layers.

(41) In some embodiments, the hole accumulation layer **238** may be deposited using a conformal deposition method such as Atomic Layer Deposition (ALD), chemical vapor deposition (CVD), or the like.
(42) In some embodiments, an over-oxidization treatment is performed after deposition of the hole accumulation layer **238** to increase interstitial oxygen (Oi) in the hole accumulation layer **238**. In some embodiments, the over-oxidization treatment may be performed by exposing the hole accumulation layer **238** to plasma of an oxygen source, such as nitrogen oxide (N.sub.2O), ozone (O.sub.3), and the like, for a period of time. In some embodiments, the over-oxidization treatment may be performed between 60 seconds and 300 seconds. In some embodiments, the over-oxidization treatment is performed at a temperature lower than 410° C., for example between 300° C. and 400° C.
(43) In some embodiments, after the over-oxidization treatment, the interstitial oxygen (Oi) in the hole accumulation layer **238** is greater than about 7% of bulk oxygen in the hole accumulation layer **238**. Bulk oxygen refers to oxygen in a bond with metal atom in a metal oxide. In some embodiments, the ratio of interstitial oxygen over bulk oxygen may be in a range between about 7% and about 12%. Not meant to bound by theory, it has been observed that metal oxide material with a higher areal oxygen density has a larger flat band voltage, thus enable strong “electron” capture and hole accumulation in semiconductors, such as silicon. Additional interstitial oxygen in the hole accumulation layer **238** further increases the areal hydrogen density, thus, improving hole accumulation at the interface **236***i* with the semiconductor substrate **202**. When the ratio of interstitial oxygen over bulk oxygen is less than 7%, the interstitial oxygen may not produce meaningful improvement to hole accumulation. When the ratio of interstitial oxygen over bulk oxygen is greater than 12%, the hole accumulation layer **238** may become unstable or lose structural integrity.
(44) In some embodiments, the hole accumulation layer **238** has a thickness T**4** (shown in FIG. **10**A) in a range between about 50 angstroms and 500 angstroms. If the thickness of the hole accumulation layer **238** is less than about 50 angstroms, the hole accumulation layer **238** may not provide adequate ability of hole generation at the interface **236***i* with the semiconductor substrate **202**. A thickness greater than about 500 angstroms may not provide additional benefit. In some embodiments, a ratio of the thickness of the defect repairing layer **236** over the thickness of the hole accumulation layer **238** is in a range between 0.01 and 1.0.
(45) In some embodiments, the high-k material in the hole accumulation layer **238** may have advantageously optical reflective properties to provide optical isolation between the pixel elements **222**.
(46) In operation **114**, a filing material **240** is deposited on the hole accumulation layer **238** and fill the deep isolation trenches **234**, as shown in FIG. **9**. In some embodiments, the filling material **240** may be a dielectric material. For example, the filling material **240** may be an oxide, such as silicon oxide. The filling material **240** may be deposited using any suitable process, such as chemical vapor deposition (CVD), atomic layer deposition (ALD). In some embodiments, the silicon oxide is formed by CVD using a suitable precursor, such as silane (SiH.sub.4) or tetraethoxysilane or Si(OC.sub.2H.sub.5).sub.4 (TEOS). In some embodiments, the filling material **240** may over fill the deep isolation trenches **234** after deposition.
(47) In some embodiments, the filling material **240** may be material with high optical reflective properties, such as a material with higher than about 90 percent reflectivity to wavelength greater than about 600 nm. In some embodiments, the high reflective material may be a metallic material, such as copper, and aluminum copper (AlCu). In some embodiment, the high reflective material may be formed by performing a physical vapor deposition (PVD) to form a seed layer followed by a plating process to fill the deep isolation trenches **234** with the metallic material.
(48) In some embodiments, air gaps may be present after deposition of the filling material **240**. For example, air gaps may be present within the filling material **240** in the deep isolation trenches **234**.

In other embodiments, the filling material **240** may choke off the deep isolation trench **234** leaving portions of the hole accumulation layers **238** exposed to the air gaps.
(49) In some embodiments, one or more air gaps may be formed in the filling material **240**. In some embodiments, a subsequent planarization process, such as a CMP process, may be performed to expose the doped regions **206**.
(50) The filing material **240**, the hole accumulation layer **238**, and the defect repairing layer **236** in the deep isolation trenches **234** form a backside deep trench isolation (BDTI) structure **242**. FIG. **9**A is a schematic top view of the semiconductor device **200** after operation **114**. FIG. **9**A illustrates that the BDTI structure **242** forms a BDTI grid surrounding the plurality of doped regions **206**. Each doped region **206** corresponds to one pixel element **222** and functions as the light sensing area in the pixel element **222**. The BDTI structure **242** provides optical and electrical isolation to the plurality of doped regions **206**. In some embodiments, the defect repairing layer **236** and the hole accumulation layer **238** function to provide passivation to the adjacent pixel element **222**. The defect repairing layer **236** and the hole accumulation layer **238** may be referred to as a composite passivation layer.
(51) In operation **116**, a plurality of color filters **246** are formed over the plurality of device elements **222**, as shown in FIG. **10**. In some embodiments, one or more absorption enhancement layers **244** may be deposited on the back surface **206***b* prior to forming the color filters **246**. The one or more absorption enhancement layers **244** is configured to increase absorption of radiation by the doped regions **206** by providing for a low reflection of radiation from the semiconductor substrate **202**. In some embodiments, the one or more absorption enhancement layers **244** may comprise a high-k dielectric material and a layer of silicon oxide.
(52) The plurality of color filters **246** are formed over the one or more absorption enhancement layers **244**. In some embodiments, the plurality of color filters **246** are aligned with the plurality of device elements **222**. The plurality of color filters **246** may be formed by depositing a color filter layer and patterning the color filter layer. The color filters **246** are formed of material that allows transmission of radiation having a specific range of wavelength while blocking light of wavelengths outside of the specified range.
(53) In some embodiments, an isolation structure **248** may be formed between neighboring color filters **246** to prevent radiation transmitted from one color filters **246** from projecting into the doped regions **206** under neighboring color filters **246**.
(54) In operation **118**, a plurality of micro lenses **250** are formed over the plurality of color filters **246**, as shown in FIG. **10**. In some embodiments, the plurality of micro lenses **250** may be formed by depositing a micro lens material above the plurality of color filters **246** by a suitable process, such as a spin-on method or a deposition process. In some embodiments, a micro lens template having a curved upper surface is patterned above the micro lens material. The micro lens template may comprise a photoresist material, for example, for a negative photoresist. For a negative photoresist, more light is exposed at a bottom of the curvature and less light is exposed at a top of the curvature. The micro lens template is then developed and baked to form a rounding shape. The plurality of micro lenses **250** are then formed by selectively etching the micro lens material according to the micro lens template.
(55) As shown in FIG. **10**, the semiconductor device **200** is an imaging sensing device including a plurality of device elements **222** with the doped regions **206** as light sensing regions separated by the BDTI structure **242**. Electromagnetic radiation **252**, such as light, projects to the doped regions **206** through the corresponding micro lens **250** and color filter **254**. As shown in FIG. **10**, the BDTI structure **242** functions as optical isolator preventing electromagnetic radiation **252** from entering neighboring doped regions **206**.
(56) Upon receiving the electromagnetic radiation **252**, the doped regions **206** emit electrons due to the photoelectric effect. The BDTI structure **242** also functions to prevent the electrons from migrating into neighboring doped regions. FIG. **10**A is a schematic enlarged partial view of the

semiconductor device **200** in an area marked **10**A in FIG. **10**. As shown in FIG. **10**A, the nitrogen rich or hydrogen rich defect repairing layer **236** reduces density of interface traps (Dit) at the interface **236***i*, therefore, increases hole accumulation in the DPW **216** of the semiconductor substrate **202**.
(57) In some embodiments, the defect repairing layer **236** is a nitrogen containing material, and the defect repairing layer **236** and the hole accumulation layer **238** may have a peak atomic concentration of nitrogen (N) in a range between about 1E04/cm.sup.3 and about 3E04/cm.sup.3. FIG. **11** is a schematic graph of atomic concentration of nitrogen in one example of the defect repairing layer **236** and the hole accumulation layer **238**. Curve **302** denotes the atomic concentration of nitrogen in the defect repairing layer **236** including aluminum nitride and the hole accumulation layer **238** including aluminum oxide. As shown in the curve **302**, the peak nitrogen concentration occurs near the interface between the defect repairing layer **236** and the hole accumulation layer **238**. Curve **304** denotes the atomic concentration of nitrogen in an aluminum oxide layer directly formed on the semiconductor layer as hole accumulation layer. As shown in FIG. **11**, the atomic concentration of nitrogen according to the present disclosure is about 2 orders higher than state-of-art hole accumulation layer.
(58) Referring back to FIG. **10**A, the hole accumulation layer **238** according to the present disclosure is formed with oxide having an areal oxygen density greater than areal oxygen density of silicon oxide. In some embodiments, a ratio of the areal oxygen density in the hole accumulation layer **238** over the areal oxygen density in silicon oxide is in a range greater than 1.0 and less than 1.4. In some embodiments, the hole accumulation layer **238** also include interstitial oxygen at a ratio between 7% and 12% of bulk oxygen. FIG. **12** is a schematic X-ray photoelectron spectroscopy (XPS) spectrum of oxygen **1***s* of the hole accumulation layer **238**. A main peak **306** is for oxygen with a lower binding energy (such as 531 eV), which corresponds to lattice oxygen or bulk oxygen, such as oxygen in Metal-O covalence. A minor peak **308** is for oxygen with a higher binding energy (such as 533 eV), which corresponds to non-lattice oxygen, such as interstitial oxygen. With an increase in the minor peak, the flat band voltage shifts positively, resulting in stronger electron capture power in the hole accumulation layer **238**. In some embodiments, the flat band voltage may increase more than 8%. For example, the flat band voltage may increase from 1.94 volt to 2.11 volt with more negatively charge.
(59) It has been observed that a composite passivation layer including AlN and AlO according to the present disclosure has density of interface trap in a range between 6.4E+11 and 4.2E+11, which is about 34% improvement over current technology. In one example, the composite passivation layer according to the present disclosure includes an AlO layer as the hole accumulation layer **238** and an AlN layer with a thickness of 5 nm as the defect repairing layer **236**. The density of interface trap on AlO-silicon interface is lowered to about 3E+11. The density of interface trap on AlO-silicon interface without the AlN layer is about 1E+12. Therefore, the composite passivation layer according to the present disclosure improves the density of interface trap by about 67%.
(60) Various embodiments or examples described herein offer multiple advantages over the state-of-art technology. By using a nitrogen or a hydrogen rich layer between the semiconductor substrate and the high-k hole accumulation layer to passivate dangling bonds in a semiconductor substrate, the DTI structure according to the present disclosure may be used to effectively provide electrical isolation without high temperature anneal.
(61) Some embodiments of the present disclosure provide a method. The method comprises forming a plurality of pixel elements in and on a front side of a semiconductor substrate; forming deep isolation trenches from a backside of the semiconductor substrate, wherein the deep isolation trenches surround a plurality of doped regions corresponding to the plurality of pixel elements; depositing a defect repairing layer on sidewalls of the deep isolation trenches, wherein the defect repairing layer containing nitrogen or hydrogen; depositing a hole accumulation layer on the defect repairing layer, wherein the hole accumulation layer comprises a metal oxide; over-oxidizing the

hole accumulation layer to add interstitial oxygen to the hole accumulation layer; filling the deep isolation trenches with a filling material; forming a plurality of color filters; and forming a plurality of micro lenses.
(62) Some embodiments of the present disclosure provide a method. The method comprises doping a semiconductor substrate to form a doped region with a first dopant; doping a gap region around the doped region with a second dopant; forming a pixel element over the doped region and the gap region from a front surface of the semiconductor substrate; forming an interconnect structure over the pixel element; etching a deep isolation trench in the gap region from a back surface of the semiconductor substrate; depositing a nitrogen containing layer on sidewalls of the deep isolation trench, wherein the nitrogen containing layer has a peak atomic concentration of nitrogen in a range between about 1E04/cm.sup.3 and about 3E04/cm.sup.3; depositing a high-k dielectric layer on the nitrogen containing layer; depositing a filling material on the high-k dielectric layer to fill the deep isolation trench; forming a color filter on the back surface of the semiconductor substrate; and forming a micro lens on the color filter.
(63) Some embodiments of the present disclosure provide a structure. The structure comprises a plurality of pixel elements formed in and on a semiconductor substrate; a deep trench isolation (DTI) structure formed in the semiconductor substrate, wherein the DTI structure separates individual pixel elements, and the DTI structure comprises: a defect repairing layer in contact with the semiconductor substrate, wherein the defect repairing layer contains nitrogen or hydrogen; a hole accumulation layer in contact with the defect repairing layer, wherein the hole accumulation layer comprises a metal oxide, and a ratio of interstitial oxygen over bulk oxygen is greater than 7%; and a filling material in contact with the hole accumulation layer.
(64) The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A method comprising: forming a plurality of pixel elements in and on a front side of a semiconductor substrate; forming deep isolation trenches from a backside of the semiconductor substrate, wherein the deep isolation trenches surround a plurality of doped regions corresponding to the plurality of pixel elements; depositing a defect repairing layer on sidewalls of the deep isolation trenches, wherein the defect repairing layer comprises a metal hydroxide; depositing a hole accumulation layer on the defect repairing layer, wherein the hole accumulation layer comprises a metal oxide; over-oxidizing the hole accumulation layer to add interstitial oxygen to the hole accumulation layer; filling the deep isolation trenches with a filling material; forming a plurality of color filters; and forming a plurality of micro lenses.
2. The method of claim 1, wherein the defect repairing layer comprises aluminum hydroxide (AlOH).
3. The method of claim 2, wherein the defect repairing layer has a thickness in a range between about 1 angstrom and about 50 nm.
4. The method of claim 1, wherein depositing the hole accumulation layer comprises depositing a high-k dielectric layer.
5. The method of claim 4, wherein over-oxidizing the hole accumulation layer comprises treating the hole accumulation layer with a plasma of oxygen source at a temperature below about 410° C.

6. The method of claim 5, wherein a ratio of interstitial oxygen over bulk oxygen in the hole accumulation layer is greater than 7%.
7. The method of claim 5, wherein the hole accumulation layer has a thickness in a range between about 50 angstroms and 500 angstroms.
8. The method of claim 1, wherein a density of defect traps at an interface between the defect repairing layer and the semiconductor substrate in a range between 3.6E11 and about 4.2E11.
9. A method, comprising: doping a semiconductor substrate to form a doped region with a first dopant; doping a gap region around the doped region with a second dopant; forming a pixel element over the doped region and the gap region from a front surface of the semiconductor substrate; forming an interconnect structure over the pixel element; etching a deep isolation trench in the gap region from a back surface of the semiconductor substrate; passivating dangling semiconductor bonds on sidewalls of the deep isolation trench by depositing a defect repairing layer on the sidewalls of the deep isolation trench; depositing a high-k dielectric layer on the defect repairing layer; treating the high-k dielectric layer with an oxygen source to increase interstitial oxygen in the high-k dielectric layer; depositing a filling material on the high-k dielectric layer to fill the deep isolation trench; performing a planarization process to remove the filing material, the high-k dielectric layer and the defect repairing layer from the back surface of the substrate; forming a color filter on the back surface of the semiconductor substrate; and forming a micro lens on the color filter.
10. The method of claim 9, wherein a ratio of interstitial oxygen over bulk oxygen in the high-k dielectric layer is in a range between 7% and 12%.
11. The method of claim 10, wherein a density of defect traps on an interface between the defect repairing layer and the gap region of the semiconductor substrate in a range between 3.6E11 and about 4.2E11.
12. A structure, comprising: a plurality of pixel elements formed in and on a semiconductor substrate; a deep trench isolation (DTI) structure formed in the semiconductor substrate, wherein the DTI structure separates individual pixel elements, and the DTI structure comprises: a defect repairing layer in contact with the semiconductor substrate, wherein the defect repairing layer contains hydrogen, and the defect repairing layer contains a metal hydroxide; a hole accumulation layer in contact with the defect repairing layer, wherein the hole accumulation layer comprises a high-k dielectric material having an areal oxygen density greater than an areal oxygen density of silicon oxide, and a filling material in contact with the hole accumulation layer.
13. The structure of claim 12, wherein the hole accumulation layer comprises a metal oxide, and a ratio of interstitial oxygen over bulk oxygen is greater than 7%.
14. The structure of claim 13, wherein the ratio of interstitial oxygen over bulk oxygen in the hole accumulation layer is less than 12%.
15. The structure of claim 12, wherein the defect repairing layer comprises one of aluminum hydroxide (AlOH).
16. The structure of claim 15, wherein the defect repairing layer has a thickness in a range between about 1 angstrom and about 50 nm.
17. The structure of claim 12, wherein the deep trench isolation structure is formed in a back surface of the semiconductor substrate, and an absorption enhancement layer is formed on the back surface of the semiconductor substrate and the deep trench isolation structure.
18. The structure of claim 12, wherein the defect repairing layer has a first thickness, the hole accumulation layer has a second thickness, a ratio of the first thickness over the second thickness is in a range between about 0.01 and about 1.0.
19. The method of claim 9, wherein depositing the defect repairing layer comprises performing a low temperature deposition at a temperature lower than about 400° C.
20. The method of claim 9, wherein treating the high-k dielectric layer comprises treating the high-

k dielectric layer with a plasma of oxygen source at a temperature range between about 300° C. and about 400° C.